

Tomás, 22 · unhelmeted motorcycle passenger · raccoon eyes, Battle's sign, clear nasal discharge · anosmia on the left · **basal skull fracture** on CT

BIO 004 · WEEK 2 · CASE DEEP DIVE

The Skull

A motorcycle crash, two black eyes, and a clear fluid that should never be in a nostril.

PATIENT CASE

Tomás is 22, the unhelmeted passenger on a friend's motorcycle that lost control at low speed about an hour ago. He arrives by ambulance, awake but confused, with a Glasgow Coma Scale of 13.

On exam the trauma team notes bruising in a ring around **both eyes ("raccoon eyes")**, bruising over the **mastoid behind the right ear ("Battle's sign")**, and a slow trickle of **clear fluid from his right nostril**. When asked to identify the smell of coffee, he reports nothing on the **left**. CT of the head shows fractures crossing the floors of the **anterior and middle cranial fossae**.

Each of those four findings points at a specific bone, a specific foramen, or a specific space. The skull is the most named region in human anatomy because every bump and hole has a clinical consequence. The team needs to know which structures are at risk before deciding what to image next.

GUIDING QUESTION

What bones form the floor of the skull, what holes pass through that floor, and how do nasal CSF leak, anosmia, and bruising around the eyes and ears all point at the same fracture?

PREWORK

Companion to your Week 2 Skull prework page. Terms, video, and spaced recall for cranial bones, facial bones, sutures and fontanelles, cavities and special features, and major foramina live there.

The skull, two functional jobs

The skull has two anatomic regions that do two different jobs. The **cranium** (8 cranial bones) surrounds and protects the brain. The **face** (14 facial bones) supports the eyes, the airway entrance, the mouth, and the muscles of expression. Together, 22 bones, with one mobile joint (the temporomandibular joint between the mandible and the temporal bone).

The interior floor of the cranium is divided into three stepped levels: **anterior cranial fossa** (frontal lobes sit on it), **middle cranial fossa** (temporal lobes sit on it), and **posterior cranial fossa** (cerebellum and brainstem sit on it). Tomás's fractures cross the anterior and middle floors, which is why his findings cluster around the eyes, nose, and ear.

The eight that hold the brain



Frontal (1)

Forehead and floor of the anterior cranial fossa.
Contains the frontal sinus.



Parietal (2)

Most of the roof and upper sides of the cranial vault. Joined at the sagittal suture.



Temporal (2)

Sides and base. Houses the ear (external auditory canal, middle ear ossicles, inner ear). The mastoid process is the bony bump behind the ear.



Occipital (1)

Back of the skull and posterior fossa floor.
Contains the foramen magnum.



Sphenoid (1)

Butterfly-shaped bone at the center of the cranial base. Articulates with every other cranial bone.
Houses the sella turcica (pituitary).



Ethmoid (1)

Between the eye sockets, behind the nose.
Contributes to the nasal cavity, orbit, and anterior cranial fossa floor (cribriform plate).

The fourteen that build the face

● Maxilla (2)

Upper jaw. Holds upper teeth, forms the floor of the orbit and most of the hard palate.

● Zygomatic (2)

Cheekbones. Form the lateral wall of the orbit.

● Nasal (2)

Bridge of the nose.

● Lacrimal (2)

Tiny bones in the medial orbital wall. Hold the nasolacrimal duct.

● Palatine (2)

Posterior part of the hard palate and small contribution to the nasal cavity and orbit.

● Inferior nasal conchae (2)

Curled bones in the lateral nasal wall. Warm and humidify inspired air.

● Vomer (1)

Inferior part of the nasal septum.

● Mandible (1)

Lower jaw. The only mobile bone of the skull.

Joints between cranial bones

Cranial bones meet at **sutures**, immobile fibrous joints. The four main sutures: *coronal* (between frontal and parietals), *sagittal* (between the two parietals), *lambdoid* (between parietals and occipital), and *squamous* (between parietal and temporal).

In infants, the sutures are not yet fused and there are larger soft spots between them called **fontanelles**. The *anterior fontanelle* (diamond shaped, top of the head) closes by 18 to 24 months. The *posterior fontanelle* (smaller, back) closes by 2 to 3 months. Fontanelles let the skull deform during birth and accommodate brain growth. A sunken anterior fontanelle in an infant suggests dehydration; a bulging one suggests raised intracranial pressure.

Spaces inside the skull, and the air-filled sinuses

Within the skull are the **cranial cavity** (holds the brain), the **nasal cavity** (air conditioning for the airway), and the **orbits** (eye sockets). Embedded in several cranial and facial bones are **paranasal sinuses**, air-filled spaces lined by mucous membrane: *frontal*, *ethmoid (multiple small cells)*, *sphenoid*, *maxillary*. Sinuses lighten the skull, warm and humidify air, and provide resonance for voice. They also collect mucus when inflamed (sinusitis).

The **cribriform plate** of the ethmoid forms most of the anterior cranial fossa floor. It is perforated by tiny holes through which olfactory nerve fibers (CN I) pass from the nasal mucosa to the olfactory bulbs sitting on top. The cribriform plate is paper-thin and is the most common site of CSF leak after head trauma.

The holes that matter clinically

FORAMEN	BONE / FOSSA	WHAT PASSES THROUGH
Cribriform plate (multiple)	ethmoid / anterior fossa	CN I (olfactory)
Optic canal	sphenoid / middle fossa	CN II (optic), ophthalmic artery
Superior orbital fissure	sphenoid / middle fossa	CN III, IV, V1, VI
Foramen rotundum	sphenoid / middle fossa	CN V2 (maxillary)
Foramen ovale	sphenoid / middle fossa	CN V3 (mandibular)
Foramen spinosum	sphenoid / middle fossa	middle meningeal artery
Internal acoustic meatus	temporal / posterior fossa	CN VII, VIII
Jugular foramen	temporal-occipital / posterior fossa	CN IX, X, XI, internal jugular vein
Hypoglossal canal	occipital / posterior fossa	CN XII
Foramen magnum	occipital / posterior fossa	spinal cord / medulla, vertebral arteries, CN XI rootlets

Why this list matters at the bedside. A fracture across a fossa floor doesn't just break bone; it can shear the cranial nerves passing through. Loss of smell after head trauma (Tomás's left-sided anosmia) means the olfactory fibers passing through the cribriform plate were torn.



DRAW ZONE 1 -- Sketch the skull from the side. Label frontal, parietal, temporal, occipital, sphenoid, maxilla, zygomatic, mandible. Mark coronal, sagittal, lambdoid sutures.

Use Fine for the suture lines. They are thin zig-zag joints, not straight.



DRAW ZONE 2 -- Sketch the floor of the cranial cavity looking down from above. Label the three fossae and mark the foramina that injured Tomás.

Star the cribriform plate (anosmia) and the route of CSF leak through the nose.

Four findings, one fracture line

Each finding on the trauma exam points at a specific bone or foramen along the cranial base. The fracture line tells the story.

"Raccoon eyes (periorbital bruising)"

Blood from the **anterior cranial fossa** floor (orbital plate of frontal, cribriform plate) tracks anteriorly into the loose periorbital tissue. The bruising appears 12 to 48 hours later and is bilateral, distinguishing it from a direct orbital blow.

"Battle's sign (mastoid bruising)"

Blood from a **temporal bone** fracture tracks posteriorly to the mastoid. Suggests middle cranial fossa floor involvement.

"Clear fluid from the nostril"

CSF rhinorrhea. Tear in the dura and cribriform plate allows CSF to drip into the nasal cavity. Can be confirmed by testing the fluid for beta-2-transferrin.

"Anosmia on the left"

The olfactory nerve fibers (CN I) on the left side were sheared as they passed through the cribriform plate. Often permanent.

"Risk we have not seen yet"

Meningitis. The torn dura is a direct route from the nasal flora into the subarachnoid space. Tomás needs neurosurgical follow-up and observation.

"Why no orbital fracture findings"

Periorbital bruising came from above (anterior fossa floor), not from a direct blow to the orbit. The orbital rim is intact on CT.

Bone, foramen, nerve, vessel. A skull base injury can be unpacked one finding at a time once you know what passes through where. When you go back to your prework, the cranial nerve list will land differently — each one is now attached to a hole in a bone in a patient who is still in the ED.

